

Multimodal Prompting

Generative AI

Module 1 – Lesson 5

Today's activities

- Introduction to multimodal applications
- Multimodal LLMs
- Prompting MLLMs
- Multimodal use cases



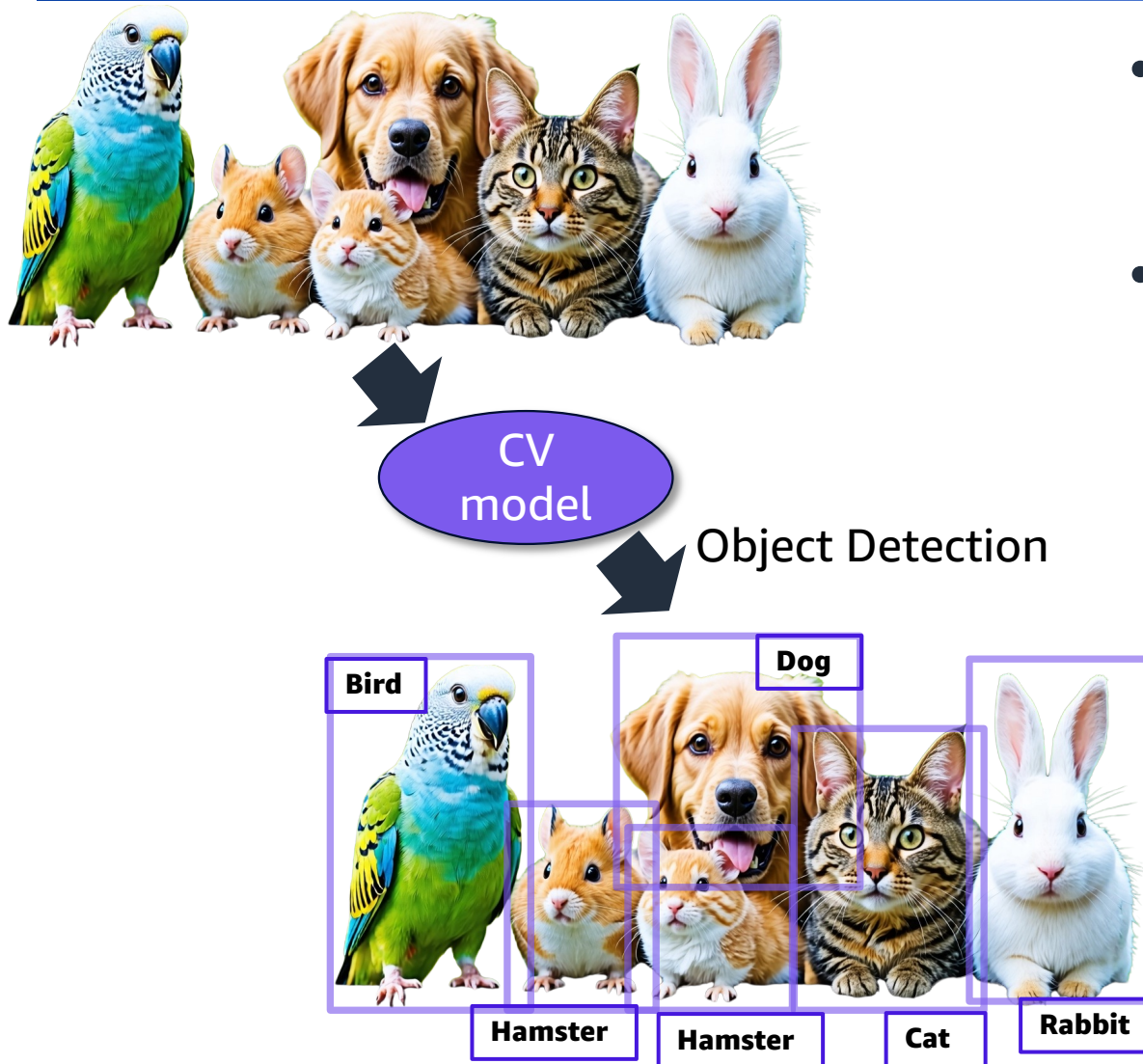
Introduction to multimodal applications

Your marketing company has been hired by a pet shop:

- They need to create individual flyers about each pet based on their images and bio
- How can you do that with traditional, single modality models?



We can consider an image-only model

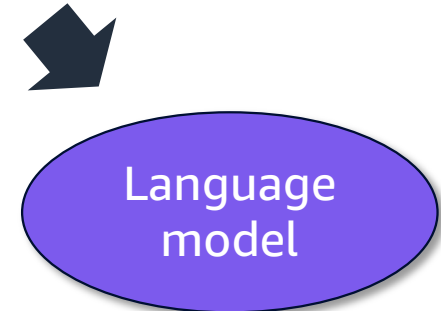


- Learns information only from images
 - Computer vision (CV)
- Can accomplish basic image understanding tasks
 - Classifying images for our campaigns depending on the theme needed
 - Ex. Cat class, dog class
 - Object detection
 - Semantic segmentation
- It won't accept the input search query or won't be able to generate text

We can consider a text-only model

- Learns information only from text
- Can accomplish basic text understanding tasks
 - Generating text based in a campaign description
- Won't be able to generate a description of the pets from images

There is a 2-year-old pomeranian looking...



for a loving home ...

What does “multimodal” mean?

- Humans are naturally multimodal in the way we interact with the world!
- Perceive the world using multiple senses:
 - Vision, hearing, smell, taste and touch
- Engage in non-verbal communication
 - Gestures
 - Facial expressions
 - Body language
 - Eye contact
 - Appearance

Why multimodal?

- Generative AI shifted from **prediction** to **interaction**
- **Multimodality** is a way to boost AI performance to interact with humans to solve real world problems

Data modalities

- **Image Data:**
 - The most versatile format for model inputs
 - There's much more visual data than text data
 - Phones and webcams constantly take pictures and videos today
 - It can be used to represent:
 - Text
 - Tabular data
 - Audio
 - And to some extent, videos

Data modalities

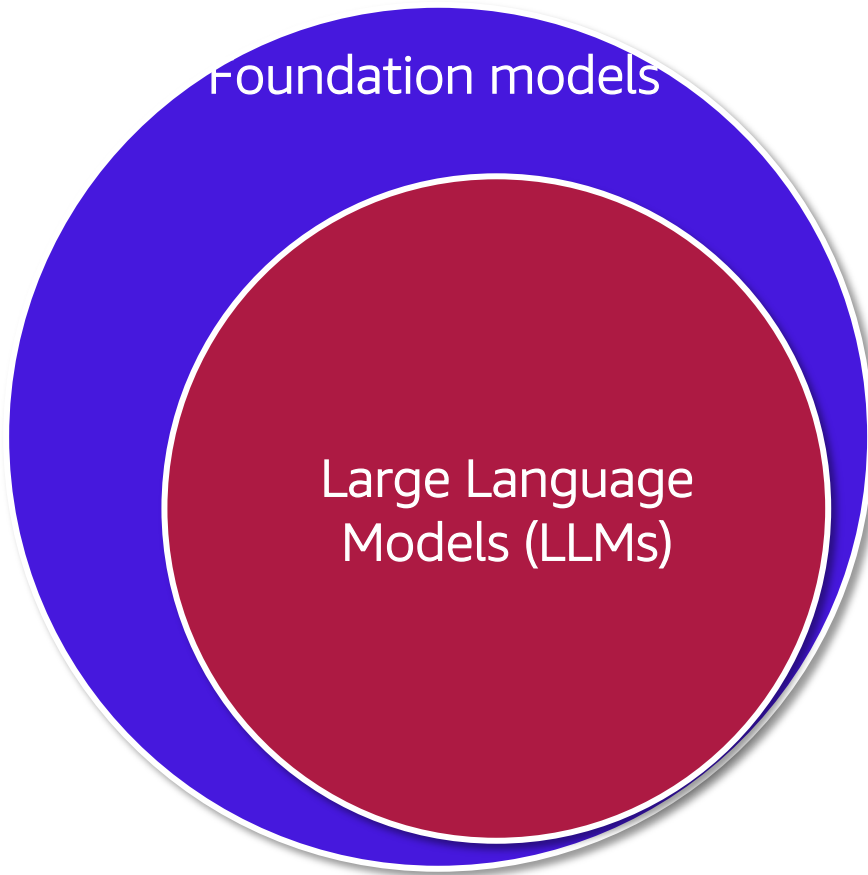
- **Text Data:**
 - Text is a powerful mode for model outputs
 - A model that can understand/generate text can be used for many tasks:
 - Summarization
 - Translation
 - Reasoning
 - question answering
 - etc.

Other data modalities

- Video
 - Audio
 - Haptic data
 - Electrical signals
-
- In this course, we will focus mainly on **text** and **image** data.

Multimodal LLMs

Review: Large Language Models (LLMs)

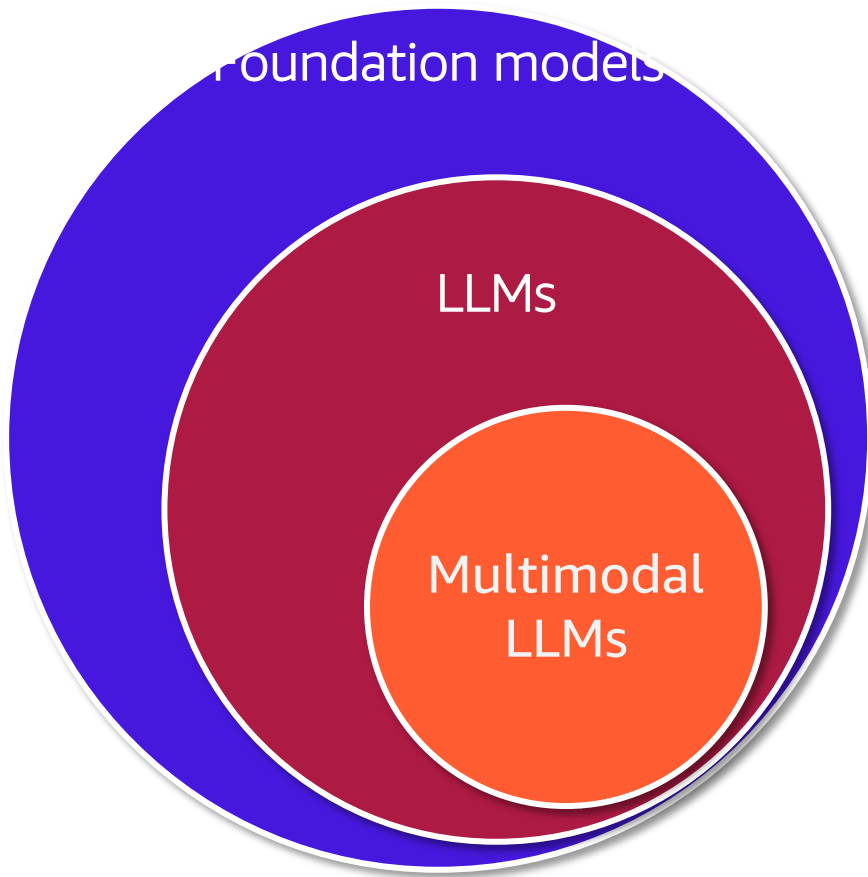


- Foundation models trained on **text**
- Large ML models that learn the **probabilities of words** being used in certain contexts
- **Training task:** Learn to predict the missing word in a text sequence

"The weather has been cloudy for the last two days. Most likely it will be ____ tomorrow."

cloudy? sunny? foggy?

Multimodal LLMs (MLLMs)



- Large language models trained on **multiple modalities**
- MLLMs typically use encoders and adapters to Equip LLMs with cross modal capabilities
 - Vision encoder
 - Video encoder
 - Audio encoder

Prompting MLLMs

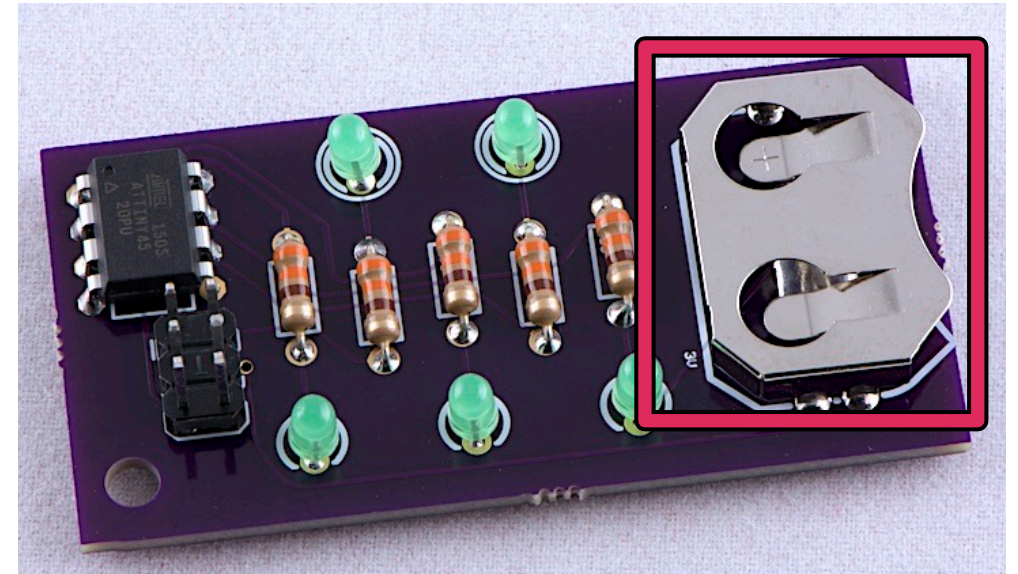
Prompting MLLMs

- **Text** prompts:
 - Follow best prompting strategies discussed in previous lessons
- **Image** prompts:
 - **Input format:** Most MLLMs use base64-encoded format
 - **Image size:** Adhere to the image size limitations (e.g. <5MB)
 - **Multiple images:** Most MLLMs can only analyze a limited number of images
 - **Image format:** Follow the image format specified for the MLLM (e.g. jpg, png, etc.)
 - **Image clarity:** Avoid blurry images
 - **Image placement:** In most cases, it works better when images come before text
 - **Image resolution:** Be within the image resolution limits of the MLLM

Multimodal use cases

Visual question answering

- Instead of relying only on text for the context, you can give the model both text and images
- **Examples:**
 - Generate text descriptions of images
 - Query using both text and images
 - Image analysis using text prompts



What is the purpose of the highlighted part in the circuit board?

Text-based image retrieval

- Image search matters not only for search engines but also for enterprises to be able to search through all their internal images and documents
- **Examples:**
 - Given a text query, find images whose captions/metadata are closest to this text query
 - Given a text query, find all images whose embeddings are closest to this embedding

Find chairs in stock

Can bring images with closest embeddings to the text



In-stock #: 235



In-stock #: 15

Using also image metadata

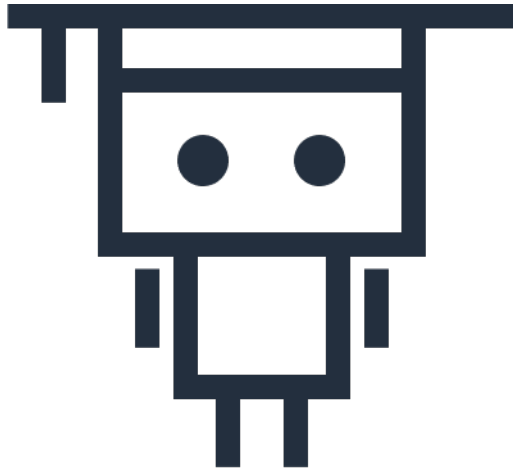
Deep image similarity retrieval

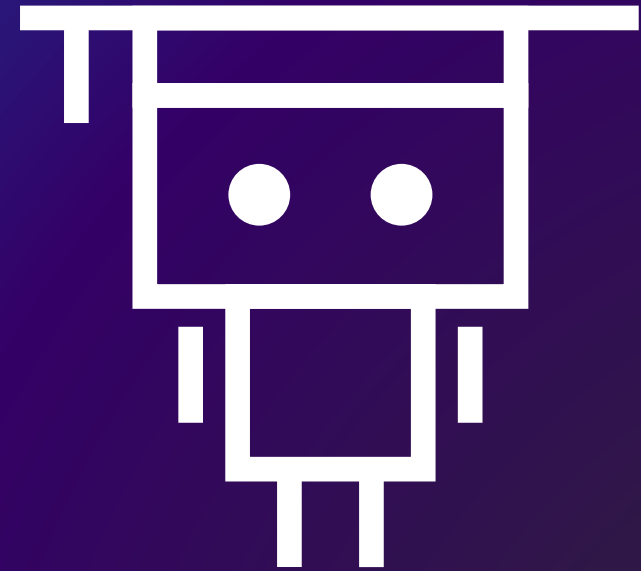
- Given an image, find similar images
- **Examples:**
 - Retrieving similar images for Amazon products
 - Identifying other product from the manufacturer



Next lesson

- This lesson introduced multimodal models and applications





Thank you!